



UTM
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SECP3213-01 KEPINTARAN PERNIAGAAN (BUSINESS INTELLIGENCE)

SECTION 1

PROJECT TITLE:

**IMPACT ANALYSIS OF GEOGRAPHIC, ECONOMIC, AND SEASONAL
DETERMINANTS ON RETAIL SALES PERFORMANCE**

PHASE 1

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1.0 Problem Background

Business Context:

In a large-scale retail operation, sales are not uniform across all locations. Performance is heavily influenced by a combination of a store's physical capacity (Store Size), the local environment (Temperature), and the regional economic health (Unemployment Rate). Additionally, these factors fluctuate significantly during Holiday periods, creating complex patterns in consumer demand.

Importance of the Problem:

Retail managers often struggle to set fair sales targets because they fail to account for the "handicaps" or "advantages" of a specific location. For example, a large store in a high-unemployment region may underperform compared to a smaller store in a stable economy. Without isolating the specific influence of temperature and holidays, the business cannot distinguish between a seasonal trend and a fundamental economic shift.

Expected Impact:

This BI solution will provide a "Normalised Sales View." By understanding how much sales typically drop when unemployment rises or temperature shifts and increase during the holiday season, stakeholders can adjust inventory and staffing levels based on predictive environmental data rather than just historical averages. It also helps business owners to estimate how large a store should be at a place when they are opening new branches, reducing the risk of underutilized space and unnecessary operational losses.

2.0 Objectives

1. To Integrate and Structure Multi-Dimensional Data:
Use Alteryx Designer to merge transactional sales records with store metadata (Size) and regional features (Temperature, Unemployment, Holidays) into a unified analytical dataset.
2. To Evaluate the Correlation Between Store Capacity and Sales:
Analyze how Store Size influences total weekly revenue to determine if larger physical locations consistently yield higher sales across different economic conditions.
3. To Measure the "Holiday Lift" Across Economic Variances:
Quantify the percentage increase in sales during Holiday weeks and determine if this lift is dampened or amplified by the regional Unemployment Rate.
4. To Assess Environmental Sensitivity:
Identify the relationship between Temperature fluctuations and sales to provide insights on weather influence on customers' shopping habits.

3.0 Dataset Description

This project uses a retail analytics dataset that includes Sales, Features, and Stores, which will be described in the following in terms of their sources, data types, and key variables .

3.1 Sales Dataset

Source

This dataset is obtained from retail data analytics on Kaggle open source.

Type of Data

This dataset is a structured data which has been organized in a tabular format consisting of rows and columns.

Key Variable / Feature

Column Name	Data Type	Description
Store	Integer	The unique identifier for each retail store indicates 1 to 45.
Dept	Integer	Department number of the store from 1 to 99.
Date	Date	Date of weekly sales record from 2010 to 2012.
Weekly__Sales	Float	Weekly total sales from the department.
IsHoliday	Boolean	Indicates whether a holiday exists within the week (True/False).

3.2 Features Dataset

Source

This dataset is gained from the same retail data analytic source on Kaggle which contains external factors that influence the sales performance.

Type of Data

This dataset is structured data which contains rows and columns in a tabular format.

Key Variable / Feature

Column Name	Data Type	Description
Store	Integer	The unique identifier for each store (1 to 45).
Date	Date	Weekly recorded date from 2010 to 2013.
Temperature	Float	Average temperature in the region.
Fuel_Price	Float	Cost of fuel in the region.
MarkDown1	Float	Promotional markdown that is only available for data after Nov 11 and not available for all stores all the time. Missing value is marked as NA.
MarkDown2	Float	Promotional markdown that is only available for data after Nov 11 and not available for all stores all the time. Missing value is marked as NA.
MarkDown3	Float	Promotional markdown that is only available for data after Nov 11 and not available for all stores all the time. Missing value is marked as NA.
MarkDown4	Float	Promotional markdown that is only available for data after Nov 11 and not available for all stores all the time. Missing value is marked as NA.
MarkDown5	Float	Promotional markdown that is only available for data after Nov 11 and not available for all stores all the time. Missing value is marked as NA.
CPI	Float	Consumer price index that indicates the economic situation.
Unemployment	Float	Unemployment rate in the region.
IsHoliday	Boolean	Indicates whether the date is in a holiday period or not (True/False).

3.3 Stores Dataset

Source

This dataset is extracted from the same retail analytics dataset on Kaggle which also includes information about the types of metadata for each store.

Type of Data

This dataset is known as structured data as it has been organized in a CSV file with columns and rows.

Key Variable / Feature

Column Name	Data Type	Description
Store	Integer	Unique identifier of each store (1 to 45).
Type	String	Category of the store (A, B, C).
Size	Integer	The physical size of the store measures in square feet.

4.0 Proposed Business Intelligence Approach

This project proposes a Business Intelligence (BI) approach that combines data preparation, integration, and visualization techniques to analyze retail sales performance under varying environmental and economic conditions. Alteryx Designer will be used for data integration, cleaning, and transformation processes, while visualization tools such as Power BI will be used to present analytical insights through interactive dashboards.

The BI workflow begins by combining weekly sales records with store information and external features data including temperature, unemployment rate, and holiday indicators. After the data is cleaned and transformed, analytical methods will be applied to identify patterns, relationships, and trends affecting retail sales performance.

4.1 Business Intelligence Tools

This project uses two tools consisting of:

- **Alteryx Designer** will be used to perform Extract, Transform, and Load (ETL) processes. It enables efficient integration of sales, store, and features datasets into a unified analytical dataset.
- **Power BI** will be used to create dashboards and visual reports that help users identify trends and compare store performance across different operational, environmental, and economic conditions.

4.2 Data Integration Process

This project integrates multiple datasets consisting of:

1. Sales
2. Store
3. Features

Using Alteryx Designer, datasets will be joined based on common attributes such as Store and Date. This integration creates a centralized dataset that allows multi-dimensional analysis across operational, environmental, and economic factors.

4.3 Data Cleaning and Transformation

The following data preparation techniques will be conducted in Alteryx Designer:

- Removal of duplicate records
- Handling missing or null values
- Standardization of date and numerical formats
- Filtering inconsistent or invalid entries
- Creation of calculated fields such as:
 - Holiday sales percentage increase
 - Average sales per store size
 - Sales variance during temperature changes

Data transformation processes such as aggregation will also be applied to support comparative and trend analysis.

4.4 Planned Data Analysis

Several analyses will be conducted to achieve the project objectives:

- **Correlation Analysis**
To evaluate the relationship between Store Size and Weekly Sales.
- **Trend Analysis**
To identify seasonal and holiday sales patterns over time.
- **Comparative Analysis**
To compare sales performance between holiday and non-holiday periods.
- **Environmental Sensitivity Analysis**
To measure how temperature changes influence customer purchasing behavior.
- **Economic Impact Analysis**
To determine whether unemployment rates weaken or strengthen sales performance during holiday periods.

4.5 Proposed Visualizations

1. Line Charts will be used to visualize weekly sales trends and identify seasonal changes.
2. Bar Charts will compare sales performance across different stores.
3. Clustered Bar Chart will compare sales performance between holiday and non-holiday seasons across different economic conditions.
4. Scatter Plots will help determine the correlation between Store Size and sales performance.
5. Heatmaps will illustrate how temperature and unemployment conditions influence sales intensity.
6. Treemap will help visualise overall sales performance across different stores.